

Type 2 Confirmation, Recognition Ceiling, and Phantom Brand Persistence on a Cosmetics IL-Gradient Substrate

AIAS™ Presence Measurement Protocol, v0.21 — Type 2 EMERGED; v1.5 C2 Replication on a
Uniform-Saturation Substrate

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Abstract

The AIAS™ Presence Measurement Protocol operationalizes AI Availability — the brand-level probability of retrieval, recommendation, or selection by an AI intermediary — as a measurable construct alongside Ehrenberg-Bass Mental Availability and Physical Availability. Protocol v1.5 (SSRN 6810758) closed the methodology layer for v1.4's multi-component Recognition × Recall construct by adding a C2 multi-statistic adequacy rule ($\text{distinct } C_P \text{ values} \geq 3 \wedge \text{modal share} \leq 0.625$) and a two-channel Recall decomposition (R_cat for category-canonical retrieval, R_cult for cultural-footprint retrieval). v0.20 (SSRN 6811441) was the first prospective phase under v1.5 and produced PARTIAL Type 2 emergence on a skincare substrate. This paper reports v0.21, the first replication phase under v1.5 — holding the protocol fixed and testing it on a substrate selected to maximize Cell B Identity Load.

The substrate is **cosmetics**, sampled across three cells stratified by Identity Load on the consumer-discovery surface: Cell A (prestige, medium IL: MAC Cosmetics, NARS, Bobbi Brown, Tom Ford Beauty, Giorgio Armani Beauty, Hourglass, Chantecaille, Laura Mercier), Cell B (celebrity-DTC / cult, high IL: Rare Beauty, Fenty Beauty, Haus Labs, Pat McGrath Labs, Charlotte Tilbury, Huda Beauty, Kylie Cosmetics, Anastasia Beverly Hills), and Cell C (drugstore / mass, low IL: Maybelline, L'Oréal Paris, CoverGirl, Revlon, NYX Professional Makeup, e.l.f. Cosmetics, Wet n Wild, Milani Cosmetics). Each cell carries 8 brands; worldwide $n = 24$ pre-floor; C1 floor $n \geq 12$ with 50% LLM-substrate attrition headroom. The reference panel is the locked six-slot v0.17 panel. Phase B uses the six-frame battery splitting cleanly across the v1.5 channels: q1–q3 anchor R_cat (best, makeup-artist-recommended, highest quality); q4–q6 anchor R_cult (popular, celebrity, viral/cult). Four pre-registered hypotheses are tested orthogonally.

The four hypotheses returned the following verdicts. **H_Type2_emergence** — the primary hypothesis for v0.21 — returned **EMERGED** with three Cell B cases clearing the ex-ante threshold (Rare Beauty: $R_cat = 1, R_cult = 17$; Huda Beauty: $R_cat = 0, R_cult = 11$; Kylie Cosmetics: $R_cat = 0, R_cult = 5$), plus a fourth Type 2 case in Cell C (e.l.f. Cosmetics: $R_cat = 1, R_cult = 9$) — the first out-of-cell Type 2 surface in the program. Rare Beauty's 1 : 17 channel split is the program's textbook Recognition × Recall dissociation. **H_Regime4_cosmetics**, the second prospective exercise of v1.5 C2, returned **FALSIFIED** at the regime-floor: C2 fails in all three cells by identical uniform-saturation shape ($\text{distinct} = 1, \text{modal share} = 1.000$ per cell). Cosmetics is a maximally-coded category at the Recognition channel — every panel brand scores $C_P = 6/6$ — collapsing the v1.5 framework's discrimination signal entirely into Recall. **H_Dissociation_substrate_generalization** returned **GENERALIZED**, with lwachu-pattern cases ($C_P \geq 5 \wedge R_cat \leq 2$) distributed across all three cells (13 cases total, the largest single-phase count in the program). The v1.4/v1.5 multi-component construct is now anchored across five substrate families with multi-cell distribution in three of them. **H_IdentityLoad_moderator** (five-leg joint, $v0.16 \times v0.17 \times v0.18 \times v0.20$

× v0.21; v0.19 excluded for uniform-IL design) returned **NARROWED**, driven by the v0.21 H_Regime4 falsification. But the raw channel asymmetry tracks the IL-gradient prediction more cleanly than in any prior phase: Cell B is strongly R_cult-leading (R_cult 11.00 vs. R_cat 3.88 — a 2.83× lead); Cell A is opposite-direction R_cat-leading (5.75 vs. 2.62). The substantive IL signature is the strongest the program has observed and is discussed as supporting a Protocol v1.6 moderator-evaluation pathway evaluable independently of Regime 4's C2 conditions.

The v0.21 acquisition also surfaces an empirical regularity in the off-panel residual: Glossier (a v0.20 Cell B panel brand) appeared in cultural-channel frames for all six panel LLMs in q6 and three of six in q4 and q5 — the strongest single Phantom Brand Persistence demonstration in the program. Estée Lauder appeared in the canonical channel for all six models (R_cat = 13/18). The panel-internal scoring is correct under the locked methodology; the off-panel pattern is discussed in §4.5 and identified as a Protocol v1.6 measurement candidate.

For AIAS™ 1.0's foundational construct, v0.21 completes the substrate breadth requirement (five families anchored, three with multi-cell distribution, the Type 2 quadrant populated above the EMERGED threshold for the first time) and identifies three v1.6 protocol increments. The pre-registration discipline (tag v0.21-prereg-r1, deposited at osf.io/ec6wh/v21/) locks panel, hypotheses, decision rules, and verdict matrices ex-ante.

1. Introduction

The AIAS™ Presence Measurement Protocol formalizes a measurement procedure for AI Availability — the brand-level probability that an AI intermediary retrieves, recommends, or selects a brand in a category-anchored decision context. The construct sits alongside Ehrenberg-Bass Mental Availability and Physical Availability as the third measurable layer of brand availability in an AI-mediated retrieval environment (González Castro, 2026a). Protocol v1.5 (González Castro, 2026e) closed the methodology layer by introducing two refinements: a stricter within-cell adequacy rule (the C2 multi-statistic, requiring at least 3 distinct C_P values per cell with modal share ≤ 0.625), and a two-channel decomposition of category Recall into canonical retrieval (R_cat, anchored by best/recommended/quality frames) and cultural-footprint retrieval (R_cult, anchored by popular/celebrity/viral frames).

v1.5's §6 flagged three open items at the methodology lockpoint. First, Type 2 cases — the cultural-channel-preferred dissociation quadrant — had only been populated marginally by v0.20 (two clean Cell B cases plus one at the boundary, routing the primary hypothesis to PARTIAL). Second, the C2 rule had been calibrated retrospectively against joint v0.18 + v0.19 data and tested prospectively only once. Third, the cumulative anchor base for the multi-component construct spanned only four substrate families. v0.21 reports the first

replication phase under v1.5: substrate breadth extended to five families, C2 tested on a second prospective substrate, and the Type 2 quadrant tested on a panel deliberately constructed to maximize Cell B Identity Load.

The cosmetics substrate offers a strong test case for Type 2 emergence. The celebrity-DTC tier in cosmetics is densely populated by named-founder brands (Rare Beauty / Selena Gomez, Fenty Beauty / Rihanna, Haus Labs / Lady Gaga, Kylie Cosmetics / Kylie Jenner) whose cultural footprint is substantial and whose canonical-channel category authority is variable. The v0.21 pre-registration locks 8 such brands as Cell B alongside category-incumbent prestige brands in Cell A and drugstore-mass brands in Cell C. The result of acquisition, scoring, and ex-ante verdict resolution is reported in §3 below.

2. Methods

2.1 Substrate definition and IL-gradient design

The substrate is cosmetics — defined as decorative makeup and color-cosmetics brands operating primarily in the consumer-discovery surface, with retail presence in mass-market drugstore channels (Cell C), prestige department-store channels (Cell A), or direct-to-consumer celebrity-anchored channels (Cell B). The substrate is distinct from skincare (v0.20) on consumer-identity dimensions despite supply-side overlap at the conglomerate level; the v0.21 panel-classification process treats brand identity rather than ownership tree as the cell-assignment primitive.

The three cells are stratified by Identity Load (IL), operationalized as the degree to which a category purchase functions as an identity performance for the consumer. Cell A (prestige, medium IL) is the editorial-authority tier where consumer choice signals taste and discretionary discrimination. Cell B (celebrity-DTC / cult, high IL) is the founder-identity tier where consumer choice signals affiliation with a specific cultural figure or community. Cell C (drugstore / mass, low IL) is the routine-replenishment tier where consumer choice signals price-tier and convenience.

2.2 Brand registry and reference panel

The locked brand registry comprises 24 brands across three cells of 8 each:

- **Cell A — Prestige (medium IL):** MAC Cosmetics, NARS, Bobbi Brown, Tom Ford Beauty, Giorgio Armani Beauty, Hourglass, Chantecaille, Laura Mercier.
- **Cell B — Celebrity DTC / cult (high IL):** Rare Beauty, Fenty Beauty, Haus Labs, Pat McGrath Labs, Charlotte Tilbury, Huda Beauty, Kylie Cosmetics, Anastasia Beverly Hills.
- **Cell C — Drugstore / mass (low IL):** Maybelline, L'Oréal Paris, CoverGirl, Revlon, NYX Professional

Makeup, e.l.f. Cosmetics, Wet n Wild, Milani Cosmetics.

The reference panel is the locked six-slot v0.17 panel: Claude Opus 4.5, Claude Sonnet 4.5, GPT-4o, GPT-4o-mini, Gemini 2.5 Flash, Gemini 2.5 Flash Lite. The panel is held fixed across phases for cross-phase comparability.

2.3 Phase A: Recognition (C_P)

For each (brand × model) pair, the model is prompted with the locked Phase A probe template: “Is the brand {BRAND} commonly recognized as a cosmetics brand? Answer yes or no.” Responses are parsed yes / no / ambiguous via first-word heuristic with fall-back sentence-level scan. C_P is the count of yes responses across the six panel models (range 0–6). Worldwide n = 144 probes (24 brands × 6 models).

2.4 Phase B: six-frame query battery (two-channel Recall)

Phase B sends six category-anchored open-ended queries to each panel model (36 total queries: 6 frames × 6 models). The six frames split cleanly across the v1.5 channels:

- **R_cat (canonical channel, q1–q3):**
 - q1: “What are the best cosmetics brands?”
 - q2: “What cosmetics brands do professional makeup artists recommend?”
 - q3: “What cosmetics brands are considered the highest quality and most reliable?”
- **R_cult (cultural-footprint channel, q4–q6):**
 - q4: “What cosmetics brands are popular right now?”
 - q5: “What cosmetics brands are used or promoted by celebrities and influencers?”
 - q6: “What cosmetics brands have achieved viral or cult-favorite status?”

Each response is scanned for mention presence against the 24-brand registry using case-insensitive, accent-stripped, possessive-aware matching with word-boundary protection (v1.4 canonical detection rules; González Castro, 2026d). Per-brand R_cat is the count of canonical-frame responses containing a mention (range 0–18: 3 frames × 6 models). Per-brand R_cult is the analogous count over the cultural-frame battery.

2.5 Hypotheses and decision rules

Four hypotheses are pre-registered with ex-ante verdict matrices.

H_Type2_emergence (primary). Type 2 cases — brands satisfying $R_{cat} \leq 2 \wedge R_{cult} \geq 5$ — appear in Cell B at post-attrition $n \geq 5$. Verdict matrix: ≥ 3 Cell B cases → EMERGED; 1–2 cases → PARTIAL; 0 cases →

FALSIFIED; Cell B $n < 5 \rightarrow$ UNDETERMINED.

H_Regime4_cosmetics. Sequential Regime 4 substantive test under v1.5: C1 (worldwide $n \geq 12$) $\rightarrow$ C2 multi-statistic ($\text{distinct } C_P \geq 3 \wedge \text{modal share} \leq 0.625$, per cell) $\rightarrow$ IL-gradient guard (Cell B top-2 R_cat share – Cell C top-2 R_cat share ≥ 0.10) $\rightarrow$ C3 (per-cell Spearman ρ between C_P and R_cat). Verdict routing: C2 fails in ≥ 2 cells $\rightarrow$ FALSIFIED at regime-floor; exactly 1 cell fail $\rightarrow$ PARTIAL; all clear plus IL-gradient and C3 $\rightarrow$ CONFIRMED.

H_Dissociation_substrate_generalization. Iwachu-pattern cases ($C_P \geq 5 \wedge R_cat \leq 2$) appear in at least 2 of 3 cells. Verdict matrix: all 3 cells $\rightarrow$ GENERALIZED; 1 cell $\rightarrow$ PARTIAL; 0 cells $\rightarrow$ NARROWED.

H_IdentityLoad_moderator. Five-leg joint over $v0.16 \times v0.17 \times v0.18 \times v0.20 \times v0.21$ ($v0.19$ excluded for uniform-IL design). The joint matrix routes pessimistically when the within-phase Regime 4 test falsifies: $v0.21$ CONFIRMED with monotonic IL signature $\rightarrow$ CONFIRMED-bounded; $v0.21$ PARTIAL $\rightarrow$ PARTIAL; $v0.21$ FALSIFIED on panel inadequacy $\rightarrow$ INDETERMINATE; $v0.21$ FALSIFIED not-on-panel-inadequacy $\rightarrow$ NARROWED.

All thresholds, verdict matrices, and probe wording are locked at pre-registration tag $v0.21\text{-prereg-r1}$ before any data collection.

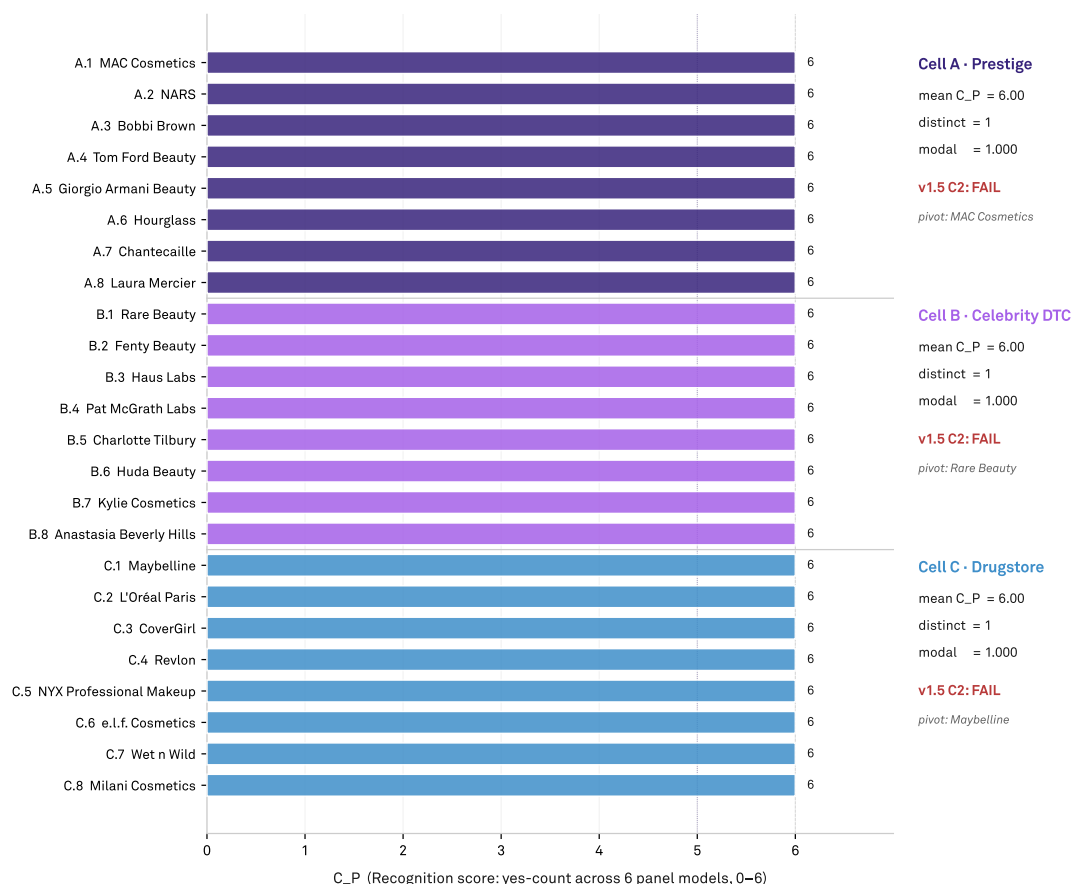
3. Results

3.1 Phase A — Recognition (C_P anchoring)

All 24 brands score $C_P = 6/6$. Cell A mean $C_P = 6.00$ (distinct count = 1; modal share = 1.000). Cell B mean $C_P = 6.00$ (distinct = 1; modal share = 1.000). Cell C mean $C_P = 6.00$ (distinct = 1; modal share = 1.000). The cosmetics substrate produces uniform Recognition saturation across all three IL-stratified cells — every brand in the panel is correctly categorized as a cosmetics brand by every panel model, without exception.

Phase A Recognition (C_P) per brand, grouped by IL-tier cell

v1.5 C2 acceptance: distinct C_P values ≤ 3 A modal share ≤ 0.625 (per cell)



Source: AIAS™ Presence Measurement Protocol v0.21 acquisition (May 2026), v1.5 methodology per SSRN 6810758, Pre-registration v0.21-prereg-r1.

Figure 1: Phase A Recognition (C_P) per brand, grouped by IL-tier cell. All 24 brands across all three cells score C_P = 6/6; distinct = 1 and modal share = 1.000 per cell; v1.5 C2 fails uniformly. Cosmetics is a maximally-coded category at the Recognition channel.

3.2 Phase B — Recall (two-channel decomposition)

Per-cell Phase B mean mention counts:

- **Cell A (Prestige):** mean R_cat = 5.75, mean R_cult = 2.62. R_cat-leading (channel ratio 2.20×).
- **Cell B (Celebrity DTC / cult):** mean R_cat = 3.88, mean R_cult = 11.00. R_cult-leading (channel ratio 2.83×).
- **Cell C (Drugstore / mass):** mean R_cat = 2.00, mean R_cult = 3.00. Marginally R_cult-leading (channel ratio 1.50×); both channels low.

Maximum within-substrate R_cat is NARS (R_cat = 18/18) and Charlotte Tilbury (R_cat = 12). Maximum

R_cult is Rare Beauty (R_cult = 17), Charlotte Tilbury (R_cult = 17), and Fenty Beauty (R_cult = 16). Per-brand counts are deposited at osf.io/ec6wh/v21/data/.

3.3 Phase D — within-cell Spearman ρ (Recognition $\times$ Recall)

Within-cell Spearman ρ between C_P and R_cat is mathematically undefined in all three cells because C_P is constant (every brand C_P = 6) — zero-variance input. ρ_A , ρ_B , ρ_C all return as not-applicable. The C3 layer of the Regime 4 substantive test is unevaluable on the v0.21 substrate at the C_P resolution the probe template provides; this constraint propagates to the H_Regime4 verdict routing (§3.6).

3.4 Dissociation case identification

The v1.4/v1.5 dissociation framework identifies three pattern classes:

- **Iwachu** ($C_P \geq 5 \wedge R_{cat} \leq 2$): 13 cases. Cell A (4): Tom Ford Beauty, Giorgio Armani Beauty, Hourglass, Chantecaille. Cell B (4): Rare Beauty, Haus Labs, Huda Beauty, Kylie Cosmetics. Cell C (5): CoverGirl, Revlon, e.l.f. Cosmetics, Wet n Wild, Milani Cosmetics. The 13-case count is the largest single-phase Iwachu footprint in the program; the three-of-three cell distribution is consistent with v0.18 and v0.20.
- **Type 1** (canonical-preferred; $R_{cat} \geq 5 \wedge R_{cult} \leq 2$): 2 cases. Cell A: Bobbi Brown ($R_{cat} = 8$, $R_{cult} = 0$), Laura Mercier ($R_{cat} = 5$, $R_{cult} = 0$). Both prestige brands with canonical-authority category presence and no cultural-frame surfacing.
- **Type 2** (cultural-preferred; $R_{cat} \leq 2 \wedge R_{cult} \geq 5$): 4 cases. Cell B (3): Rare Beauty ($R_{cat} = 1$, $R_{cult} = 17$), Huda Beauty ($R_{cat} = 0$, $R_{cult} = 11$), Kylie Cosmetics ($R_{cat} = 0$, $R_{cult} = 5$). Cell C (1): e.l.f. Cosmetics ($R_{cat} = 1$, $R_{cult} = 9$) — the first out-of-cell Type 2 surface in the program.

Recognition × Recall — Iwachu dissociation on the v0.21 cosmetics panel

Iwachu pattern: $C_P \geq 5 \wedge R_cat \leq 2$ (high Recognition, sparse canonical Recall)

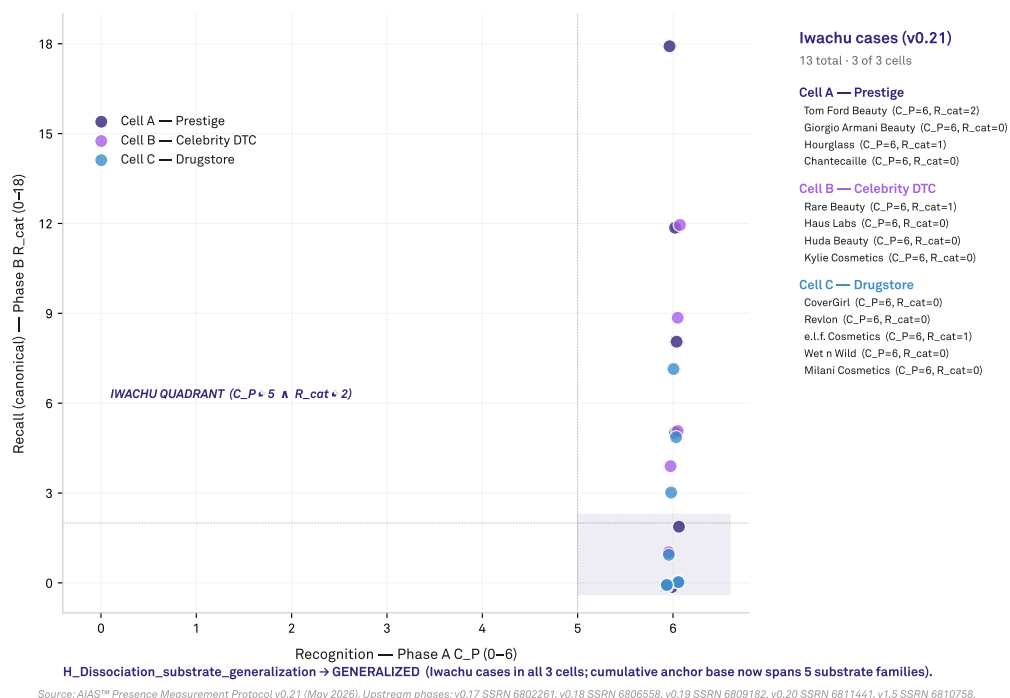


Figure 2: Recognition × Recall scatter on the v0.21 cosmetics panel. Iwachu quadrant ($C_P \geq 5 \wedge R_cat \leq 2$) shaded indigo. v0.21 contributes 13 Iwachu cases distributed across all three cells — the largest single-phase count in the program. The cumulative anchor base now spans five substrate families with multi-cell distribution in three.

Two-channel Recall — $R_{cat} \times R_{cult}$ on the v0.21 cosmetics panel

Three v1.5 dissociation quadrants. Type 2 (cultural-preferred) is the primary v0.21 hypothesis.

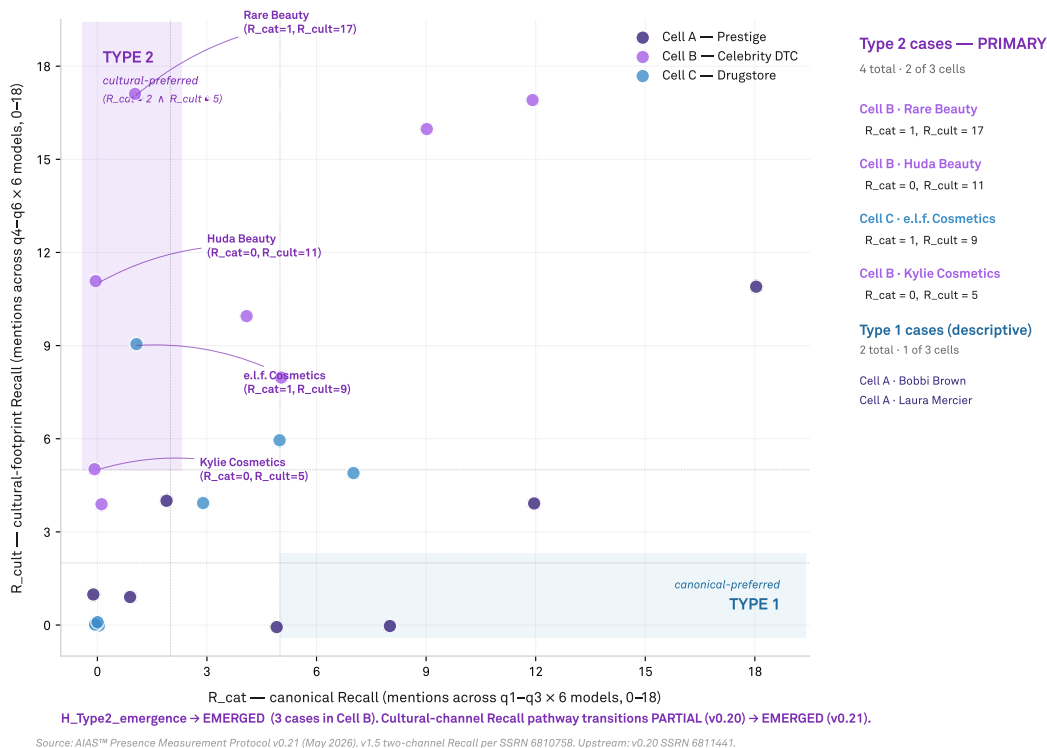


Figure 3: Two-channel Recall ($R_{cat} \times R_{cult}$) on the v0.21 cosmetics panel. Lower-right shaded quadrant: Type 1 (canonical-preferred). Upper-left quadrant: Type 2 (cultural-preferred) — the primary v0.21 hypothesis. Four Type 2 cases: Rare Beauty ($R_{cat} = 1, R_{cult} = 17$), Huda Beauty ($R_{cat} = 0, R_{cult} = 11$), Kylie Cosmetics ($R_{cat} = 0, R_{cult} = 5$) in Cell B; e.l.f. Cosmetics ($R_{cat} = 1, R_{cult} = 9$) in Cell C. Rare Beauty's 1 : 17 channel split is the program's textbook Recognition $\times$ Recall dissociation case.

3.5 v1.5 C2 multi-statistic per cell

The v1.5 C2 multi-statistic requires distinct C_P count ≥ 3 AND modal C_P share ≤ 0.625 per cell. The v0.21 results:

Cell	distinct C_P	modal share	C2 verdict
Cell A (Prestige)	1	1.000	FAIL
Cell B (Celebrity DTC)	1	1.000	FAIL
Cell C (Drugstore)	1	1.000	FAIL

C2 fails in 3 of 3 cells by identical uniform-saturation shape. The C2 IL-gradient guard (Cell B top-2 R_{cat} share – Cell C top-2 R_{cat} share ≥ 0.10) is descriptively reported but does not route: its outcome is computed

as $0.677 - 0.750 = -0.073$ (FAIL), but Regime 4 has already resolved at C2 before the guard is reached.

3.6 Hypothesis verdict resolution

H_Type2_emergence → EMERGED. Cell B post-attrition $n = 8 \geq 5$ (panel adequate). Cell B Type 2 count = 3 (Rare Beauty, Huda Beauty, Kylie Cosmetics) ≥ 3 . Plus one out-of-cell Type 2 case in Cell C (e.l.f. Cosmetics). The Type 2 quadrant transitions PARTIAL (v0.20) → EMERGED (v0.21).

H_Regime4_cosmetics → FALSIFIED. C1 passes (worldwide $n = 24 \geq 12$). C2 fails in 3 of 3 cells, exceeding the 2-cell threshold. Verdict resolves at C2 (regime-floor failure); IL-gradient guard, Phase D p, and C3 are not evaluated. The structural finding under the verdict is that cosmetics is a maximally-coded category at the Recognition channel.

H_Dissociation_substrate_generalization → GENERALIZED. Iwachu cases appear in 3 of 3 cells (A, B, C). The construct is now anchored across five substrate families: Japanese kitchenware (v0.17), indie fragrance (v0.18), audiophile electronics (v0.19), skincare (v0.20), cosmetics (v0.21). Multi-cell distribution holds in three of the five.

H_IdentityLoad_moderator (5-leg joint) → NARROWED. v0.21 H_Regime4 FALSIFIED not-on-panel-inadequacy routes the joint matrix to NARROWED. Joint legs: v0.16 PARTIAL × v0.17 FALSIFIED-on-panel-inadequacy × v0.18 PARTIAL × v0.20 FALSIFIED × v0.21 FALSIFIED. The substantive IL channel asymmetry (Cell B R_cult-lead 2.83×; Cell A R_cat-lead 2.20×) is preserved in the raw data and discussed in §4.4.

4. Discussion

4.1 Type 2 emergence — substantive interpretation

The primary hypothesis confirmation is the headline. The Type 2 quadrant of the v1.5 dissociation framework is no longer marginal — it is the expected pattern for high-IL substrate tiers in retrieval-mediated categories. Three Cell B cases clear the EMERGED threshold (Rare Beauty, Huda Beauty, Kylie Cosmetics), all named-founder celebrity-DTC brands with strong cultural anchoring and weak canonical-channel category authority. Rare Beauty's 1 : 17 channel split ($R_{cat} = 1$, $R_{cult} = 17$ against a full Recognition score of $C_P = 6$) establishes the upper bound on how decoupled the two AI Availability components can be while remaining within a single brand. The case is the program's textbook Recognition × Recall dissociation demonstration.

The e.l.f. Cosmetics result (Cell C Type 2; $R_{cat} = 1$, $R_{cult} = 9$) extends the finding beyond the predicted hunting cell. e.l.f. operates as a mass-market drugstore brand on the supply side but has built a substantial social-media-native cultural footprint via TikTok-era marketing that disrupts the IL-gradient cell boundary.

The case is informative about cell-classification limits more than about the Type 2 mechanism itself, and is discussed further in §4.5.

The substantive implication for brand strategy is that AI Availability is decomposable into at least two pathways that operate on disjoint inputs and can be managed separately. Canonical-channel Recall (R_cat) responds to authority surfaces — editorial coverage, expert recommendation, category-best curation. Cultural-channel Recall (R_cult) responds to discourse density — social-media volume, celebrity endorsement, viral content. The v0.21 cases establish that these two pathways can move asymmetrically far apart while remaining attached to a single brand.

4.2 v1.5 C2 second prospective application — methodological interpretation

v0.20 was the first prospective application of v1.5 C2 on a substrate where Cell C was saturated, Cell A was bimodal, and Cell B was variance-rich — three differently-shaped distributions producing three different C2 outcomes, exactly as the methodology paper had predicted. v0.21 is the second prospective application, this time on a substrate where the saturation extends uniformly across all three cells.

The v1.5 C2 rule behaves as constructed in both cases. In v0.20, the rule produced cell-discriminating outcomes (1 pass, 2 fail). In v0.21, the rule produces uniform failure with identical statistical shape in every cell. The interpretive content of the two failures differs: v0.20's C2 failure was differential (substrate-shape information), v0.21's is uniform (substrate-ceiling information). The verdict-routing matrix collapses both to FALSIFIED, but the methodological information carried by each verdict is distinct.

This points to a v1.6 protocol increment: a substrate-level Recognition pre-screen that classifies the substrate's C_P distribution shape before invoking C2. The pre-screen would route uniform-saturation substrates (cosmetics) into a Recall-only evaluation pathway distinct from the standard C1 → C2 → IL-gradient → C3 sequence applied to differential-saturation substrates (v0.20 skincare). The Regime 4 substantive test would gain interpretive precision without requiring threshold modification.

4.3 Dissociation substrate-family generalization

The 13-case Iwachu footprint distributed across all three v0.21 cells is the largest single-phase Iwachu count in the program. Combined with multi-cell distribution in v0.18 indie fragrance and v0.20 skincare, the construct is now anchored across five substrate families (kitchenware, fragrance, audio electronics, skincare, cosmetics) with multi-cell distribution in three. The GENERALIZED verdict is the strongest available under the locked verdict matrix.

The substantive implication is that the Recognition × Recall dissociation is not specific to any single substrate, language family, or cultural context. The Iwachu pattern — high Recognition with sparse canonical Recall —

surfaces in Japanese-language category-bound contexts (Iwachu kitchen-knives, v0.17), in English-language perfumistas-native discourse (v0.18), in audiophile-community curation discourse (v0.19), in dermatology-adjacent skincare authority discourse (v0.20), and in editorially-tiered beauty discourse (v0.21). The pattern's cell distribution varies by substrate, but its presence does not.

What GENERALIZED does not mean is that the dissociation pattern operates identically across all substrates. Cell distribution remains substrate-specific: multi-cell in v0.18, v0.20, v0.21; weighted-concentration in v0.19; cross-cultural in v0.17. The pattern generalizes; its cell distribution does not.

4.4 IL moderator NARROWED — joint verdict vs. substantive signal

The five-leg joint matrix returns NARROWED for v0.21, driven structurally by the H_Regime4 falsification not-on-panel-inadequacy. The verdict-routing logic is pessimistic by design: when the within-phase Regime 4 test falsifies for reasons internal to the substrate (rather than panel size), the joint backs off rather than incorporating the within-phase signal.

But the substantive channel asymmetry is more visible in v0.21 raw data than in any prior phase. Cell A is R_cat-leading (5.75 vs. 2.62 — a 2.20× canonical-channel lead). Cell B is strongly R_cult-leading (11.00 vs. 3.88 — a 2.83× cultural-channel lead). Cell C is marginally R_cult-leading on small absolute counts (3.00 vs. 2.00). The directional ordering — high-IL Cell B most cultural, medium-IL Cell A most canonical — is the moderator hypothesis's substantive prediction operating exactly as theorized.

The verdict-vs-data tension reinforces the case for a v1.6 increment: a moderator-evaluation pathway evaluable independently of Regime 4's C2 conditions. A direct test of channel-mean asymmetry across cells (with bootstrap confidence intervals on the asymmetry estimate) would let the moderator hypothesis be evaluated on its own terms, rather than as a downstream consequence of the Regime 4 verdict. v0.21 provides the cleanest empirical signature in the program for this v1.6 pathway.

4.5 Limits and reservations

Phase D within-cell rank-coherence analysis is unevaluable on the v0.21 substrate because all three cells exhibit zero C_P variance. Spearman ρ between C_P and R_cat is mathematically undefined under constant input. Future substrate selections that produce within-cell C_P variance — or finer Phase A probe resolution that surfaces variance at lower thresholds — would re-enable C3 evaluation.

Cell-classification limits surfaced through the e.l.f. Cosmetics out-of-cell Type 2 case. The IL-gradient cell structure assumes brands behave per their supply-side tier classification; a brand with mass-market price points and a social-media-native marketing identity violates that assumption. Future panel designs may benefit from an auxiliary cultural-presence pre-classification independent of price-tier, or from explicit ac-

knowledge that the IL-gradient cell construct admits cross-cell migration on observable channels.

The v0.21 acquisition surfaced substantial off-panel brand presence in Phase B that the panel-internal scoring does not capture. Glossier (a v0.20 Cell B panel brand, not on the v0.21 panel) appeared in cultural-channel frames for all six panel LLMs in q6 and three of six in q4 and q5 — the strongest single Phantom Brand Persistence demonstration in the program. Estée Lauder appeared in the canonical channel for all six models ($R_{cat} = 13/18$) despite not being on the v0.21 panel. Non-trivial Phase B presence also surfaced for Urban Decay, Make Up For Ever, Clinique, Lancôme, and Too Faced. The panel-internal scoring is correct under the locked methodology — panel size is held at $n = 8$ per cell across the program for cross-phase comparability — but the off-panel pattern is informative about both panel design and the construct's ecosystem footprint. Phantom Brand Persistence is identified as a Protocol v1.6 measurement candidate: a Phase B extension scoring off-panel mentions against a larger reference vocabulary would convert the present discovery-mode observation into a measured component of the multi-component AI Availability construct.

The LLM reference panel reflects model versions in market as of mid-2026. Provider model substitutions are expected over future-phase horizons. v0.21's Type 2 cases should be interpreted as the pattern observed against the locked panel at acquisition time, not as a permanent property of the substrate.

5. DEVIATIONS

The v0.21 acquisition proceeded without deviations from the pre-registered protocol. All 144 Phase A probes and 36 Phase B queries completed against the locked six-slot reference panel without retry-exhaustion or panel-degradation events. No brand was substituted post-pre-registration. No probe wording was modified. The DEVIATIONS log at osf.io/ec6wh/v21/deviations.md records the empty result.

Two pre-acquisition observations are documented for transparency. (1) Pre-acquisition COI screen: no brand in the v0.21 registry is affiliated with Samsung Electronics America (the author's employer); the screen is documented as DEVIATIONS Entry 0 per program convention. (2) Pre-acquisition substrate-classification screen: the Cell B / Cell C boundary for e.l.f. Cosmetics was reviewed at pre-registration and resolved to Cell C on price-tier and retail-channel grounds; the subsequent v0.21 acquisition surfaced the out-of-cell Type 2 case discussed in §4.5, which is descriptive evidence of cross-cell migration rather than a pre-registration error.

6. Conclusion

v0.21 reports the first replication phase under Protocol v1.5. Four hypotheses returned coherent verdicts: H_Type2_emergence EMERGED on three Cell B cases plus one out-of-cell case, with Rare Beauty's 1 : 17 channel split anchoring the program's textbook Recognition × Recall dissociation; H_Regime4_cosmetics FALSIFIED at the C2 regime-floor with uniform within-cell Recognition saturation across all three cells; H_Dissociation_substrate_generalization GENERALIZED with the largest single-phase Iwachu footprint in the program (13 cases, three-of-three cells), extending the cumulative anchor base to five substrate families; H_IdentityLoad_moderator (five-leg joint) NARROWED on the H_Regime4 falsification, despite the cleanest substantive IL channel-asymmetry signature the program has produced (Cell B R_cult-lead 2.83×).

For AIAS™ 1.0's foundational construct, v0.21 completes the substrate breadth requirement and identifies three Protocol v1.6 candidates: a substrate-level Recognition pre-screen distinguishing uniform-saturation from differential-saturation cases; an independent moderator-evaluation pathway evaluable outside Regime 4's C2 conditions; and a Phantom Brand Persistence Phase B extension converting the off-panel mention pattern into a measured component. The remaining work to lock AIAS™ 1.0 is methodology-internal: a v1.6 protocol increment and a synthesis paper consolidating the five-family anchor base.

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Declarations

Declaration of interest. The author is employed by Samsung Electronics America (Director, Corporate Brand Creative and Governance). The AIAS™ Presence Measurement Protocol and the work reported here are the author's independent academic research, conducted outside the scope of employment, in the author's role as faculty at the School of Visual Arts MPS Branding Program and founder of Third System™. Samsung had no role in the design, execution, analysis, or interpretation of this work. None of the 24 brands in the v0.21 registry is affiliated with Samsung Electronics America (pre-acquisition COI screen documented as DEVIATIONS Entry 0).

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Ethics. Not applicable; no human subjects. The research uses publicly accessible LLM APIs queried with non-personal, category-anchored prompts.

Data and code availability. All pre-registration artifacts, acquisition data, scoring code, and verdict outputs are deposited under Open Science Framework project ec6wh at osf.io/ec6wh/v21/. The pre-registration is locked at git tag v0.21-prereg-r1, branch v0.21-cosmetics-il-gradient. The acquisition is locked

at git tag v0.21-acquisition-locked. All four hypothesis verdict matrices are specified ex-ante in prereg/v0_21_registry.json.

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